

Data Storage Basics

Bits, Bytes & Megabytes



■ Learning Objectives

By the end of this lesson, students will be able to:

- ◆ Explain what a bit is and that it can only be 0 or 1.
- ◆ Describe how 8 bits combine to make 1 byte.
- ◆ Understand that a megabyte equals approximately one million bytes.
- ◆ Give real-life examples of data sizes (photos, games, videos).

■ Section 1 — What Is Data Storage?

Data storage means saving information on computers and devices. We keep our favourite photos, fun games, and exciting videos safe as data. Everything you see on a screen — text, pictures, sounds — is stored data!

■ **Key Idea: Photos, games, and videos are all saved as DATA!**

■ Section 2 — Meet the Bit

A **bit** is the tiniest piece of data a computer uses. It can only ever be a **0** (off) or a **1** (on) — like a tiny light switch. Think of one bit as one tiny Lego brick. On its own it's small, but millions of bits join together to store everything on your device!



One bit: either OFF (0) or ON (1)

■ **Fun Fact: Millions of bits work together to store every photo and song!**

■ Section 3 — What Is a Byte?

A **byte** is a group of exactly **8 bits** working together. Imagine a small box that holds 8 Lego bricks — that's a byte! When bits work together in a byte, they can store a single letter, a number, or another character. Every letter you type uses 1 byte.

Example byte — the letter 'A' (01000001 in binary):



8 bits → 1 byte → 1 character (A, B, 5, !, ...)

■ **Key Formula: 8 Bits = 1 Byte = 1 Letter or Number**

■ Section 4 — What Is a Megabyte?

A **megabyte (MB)** is roughly **one million bytes**! Think of a giant toy chest crammed with one million tiny Lego boxes. That's how enormous a megabyte really is.

Unit	Equal To	Real-World Size
1 Bit	smallest unit	A single 0 or 1
1 Byte	8 Bits	1 letter or number
1 Kilobyte	~1,000 Bytes	A short text message
1 Megabyte	~1,000,000 Bytes	A photo, short game, or clip

■ Section 5 — How Big Is a Megabyte?

Item	Approx. Size	What That Means
■ One Photo	~2 MB	A snapshot from your camera
■ Simple Game	~5 MB	A basic game on your tablet
■ Short Video	~10 MB	A 30-second video clip

■ Section 6 — Why Does Data Size Matter?

Knowing data sizes helps us make smart choices. If your tablet only has **10 MB** of free space, you need to decide: save a photo (2 MB), a game (5 MB), or a short video (10 MB)? Understanding bytes and megabytes helps you manage your device wisely. We see data sizes every day on phones, tablets, computers, and video games!

■ **Smart Tip: Knowing data sizes = making smarter choices about storage!**

■ Quick Quiz — What Do You Remember?

Q1. What is the smallest unit of data?

A: A bit — it can only be 0 or 1.

Q2. How many bits make 1 byte?

A: 8 bits = 1 byte.

Q3. Which is bigger — a byte or a megabyte?

A: A megabyte! It equals about 1,000,000 bytes.

Q4. What is one real-life example of something that uses megabytes?

A: A photo (~2 MB), a simple game (~5 MB), or a short video (~10 MB).

Q5. What can knowing data sizes help you do?

A: Make smart choices about what to save on your device.

■ Lesson Summary

■ Bits	Tiniest piece of data. Only 0 or 1.
■ Bytes	8 bits joined together. Stores 1 character.
■ Megabytes	~1,000,000 bytes. Stores photos & games.

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■ Try This at Home: With a parent or teacher, check the file sizes of photos and games on a device. Can you find something that is 2 MB? 10 MB?

■ Great Work — Keep Exploring Data Storage! ■